M and Z Estimators

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Notation for this lecture

Parameter and data. The parameter set is $\Theta \subset \mathbb{R}^p$ and the true parameter is θ_0. The observations $X_1, \ldots, X_n$ are i.i.d. copies of a random vector X of $\mathbb{R}^d$, of law $\mathcal{L}_{\theta_0}$ inside a model $\{\mathcal{L}_\theta; \theta \in \Theta\}$.

Criteria. $M_n : \Theta \to \mathbb{R}$ is a random function and $M : \Theta \to \mathbb{R}$ is its deterministic limit. For Z estimators, we use $Z_n : \Theta \to \mathbb{R}^q$ and $Z : \Theta \to \mathbb{R}^q$.

Random function means that, for every θ, $M_n(\theta)$ is a random variable, and that all the $M_n(\theta), \theta \in \Theta$, are defined on one probability space.

Subscripts. $\mathbb{E}_\theta$ is the expectation computed as if the parameter were θ, while $\mathbb{E}_X = \mathbb{E}_{\theta_0}$ is the real one. Suprema carry their index set, as in $\sup_{\theta \in \Theta}$.

Derivatives. $J\phi(\theta)$ denotes the Jacobian matrix of ϕ at θ, and $\nabla\phi(\theta)$ the gradient when ϕ is real valued.

Part I. Two new ways to define an estimator

M estimators

Definition. Let $(M_n)_{n\in\mathbb{N}}$ be a sequence of random functions from Θ to $\mathbb{R}$. An **M-estimator** is a sequence $(\hat{\theta}_n)_{n\in\mathbb{N}}$ of random vectors taking values in Θ such that, for every n,

$$\hat{\theta}_n \in \arg\max_{\theta\in\Theta} M_n(\theta) \ .$$

The letter M stands for maximizer. Since replacing M_n by $-M_n$ turns a maximization into a minimization, we lose nothing by always maximizing.

In practice, the almost sure requirement is too strong for anything computed numerically. This is why it is relaxed twice, first to holding with probability going to 1, and then to maximizing up to an additive $o_{\mathbb{P}}(1)$.

Exercise, the empirical mean

Exercise. Given random vectors $X_1, \dots, X_n$ of $\mathbb{R}^d$, express the empirical mean $\bar{X}_n = \frac{1}{n}\sum_{i\leq n} X_i$ as an M-estimator.

Solution. We take $\Theta = \mathbb{R}^d$ and

$$M_n(\theta) = -\frac{1}{n}\sum_{i\leq n} \|\theta - X_i\|^2 \ .$$

Then $\nabla M_n(\theta) = -\frac{2}{n}\sum_{i\leq n}(\theta - X_i)$ vanishes only at $\theta = \bar{X}_n$. Furthermore, since M_n is strictly concave and tends to $-\infty$ at infinity, it has a unique maximizer, which is thus $\bar{X}_n$.

Empirical risk minimization

Definition. Given a loss $\ell : \mathbb{R}^d \times \Theta \to \mathbb{R}$, the **empirical risk** and the **risk** of a parameter are

$$\widehat{R}_S(\theta) = \frac{1}{n} \sum_{i \leq n} \ell(X_i, \theta) \ ,$$
$$R(\theta) = \mathbb{E}_X [\ell(X, \theta)] \ ,$$

and the **empirical risk minimizer** is the M-estimator of $M_n = -\widehat{R}_S$, whose limit criterion is $M = -R$.

Languages from two communities. The two names come from two communities rather than from two objects, with $\ell = -m$ and $M = -R$. In supervised learning, $X = (x, y)$ and $\ell(X, \theta) = \ell(y, f_\theta(x))$ for a predictor f_θ.

Maximum likelihood

Assumption. For every θ, $\mathcal{L}_\theta$ has a density f_θ with respect to a reference measure ν.

The **likelihood** and the **log-likelihood** of the sample are $L_n(\theta) = \prod_{i \leq n} f_\theta(X_i)$ and $\log L_n(\theta) = \sum_{i \leq n} \log f_\theta(X_i)$.

Definition. The **maximum likelihood estimator** is the M-estimator associated with $M_n = \frac{1}{n} \log L_n$, that is, with

$$M_n(\theta) = \frac{1}{n} \sum_{i \leq n} \log f_\theta(X_i) \ .$$

Equivalently, it is the empirical risk minimizer for the log loss $\ell(x, \theta) = -\log f_\theta(x)$, since $M_n = -\widehat{R}_S$ for that loss.

Idea. We choose the member of the model that makes the observed sample most probable. Note that the factor $\frac{1}{n}$ does not change the maximizer, and that it makes M_n converge to something finite.

Exercise, the gaussian model

Exercise. What is the maximum likelihood estimator in the gaussian model $\Theta = \mathbb{R} \times (0, \infty), \theta = (\mu, \sigma^2), \mathcal{L}_\theta = \mathcal{N}(\mu, \sigma^2)$?

Solution. With $f_{(\mu,\sigma^2)}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$, we obtain

$$M_n(\mu, \sigma^2) = \frac{1}{n} \sum_{i \leq n} \left\{ -\log \sigma - \frac{1}{2} \log(2\pi) - \frac{(X_i - \mu)^2}{2\sigma^2} \right\} .$$

$\partial_\mu M_n = \frac{1}{n\sigma^2} \sum_{i \leq n} (X_i - \mu)$ vanishes only at $\mu = \bar{X}_n$, and

$$\partial_{\sigma^2} M_n = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} \cdot \frac{1}{n} \sum_{i \leq n} (X_i - \mu)^2 ,$$

which vanishes at $\mu = \bar{X}_n$ exactly when $\sigma^2 = \frac{1}{n} \sum_{i \leq n} (X_i - \bar{X}_n)^2$.

We recover the pair of the method of moments of the previous lecture, although the two methods differ in general.

Z estimators

Definition. Let $(Z_n)_{n\in\mathbb{N}}$ be a sequence of random functions from Θ to $\mathbb{R}^q$. A **Z-estimator** is a sequence $(\hat{\theta}_n)_{n\in\mathbb{N}}$ of random vectors taking values in Θ such that, for every n, almost surely

$$Z_n(\hat{\theta}_n) = 0 \; .$$

The letter Z stands for zero.

Every smooth M-estimator is a Z-estimator. Indeed, if Θ is open and M_n is differentiable, a maximizer in the interior satisfies $\nabla M_n(\hat{\theta}_n) = 0$, so that $\hat{\theta}_n$ is a Z-estimator with $Z_n = \nabla M_n$ and $q = p$.

The method of moments is a Z-estimator, with $Z_n(\theta) = e_n - e(\theta)$ in the notation of the previous lecture.

Part II. Consistency

The two ingredients of consistency

We want to show that $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$, a statement that mixes two things of completely different natures.

A statement about the criterion. M_n is close to M, uniformly in θ. All the randomness lies there.

A statement about the geometry. M separates θ_0 from the rest of Θ. There is no randomness in it at all.

Independence of the two statements. Each can hold while the other fails.

Stage one, the criterion error

Proposition. Let $\Delta_n = \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)|$ and assume that $\hat{\theta}_n$ nearly maximizes M_n, meaning

$$M_n(\hat{\theta}_n) \geq \sup_{\theta \in \Theta} M_n(\theta) + o_{\mathbb{P}}(1) \; .$$

Then

$$\sup_{\theta \in \Theta} M(\theta) - M(\hat{\theta}_n) \leq 2\Delta_n + o_{\mathbb{P}}(1) \; .$$

Proof. For every fixed $\theta \in \Theta$, we have

$$\begin{aligned} M(\hat{\theta}_n) &\geq M_n(\hat{\theta}_n) - \Delta_n && \text{by definition of } \Delta_n \\ &\geq M_n(\theta) - \Delta_n + o_{\mathbb{P}}(1) && \text{by near-maximization} \\ &\geq M(\theta) - 2\Delta_n + o_{\mathbb{P}}(1) \; . && \text{by definition of } \Delta_n \end{aligned}$$

Since the left side does not depend on θ, taking the supremum over θ on the right concludes the proof.

Generality of stage one

If $\Delta_n \xrightarrow{\mathbb{P}} 0$, the proposition takes the form

$$\sup_{\theta \in \Theta} M(\theta) - M(\hat{\theta}_n) \xrightarrow{\mathbb{P}} 0 \ .$$

No topology, no identifiability, no compactness, no true parameter. Θ is an arbitrary set, M is an arbitrary function on it, and we have not even used θ_0.

A statement about real numbers. The conclusion says that the value of the criterion at $\hat{\theta}_n$ approaches its optimal value, and nothing about where $\hat{\theta}_n$ is.

This is already the useful statement in many settings. In empirical risk minimization, the excess risk is this quantity, which generalization bounds control.

Empirical risk minimization as stage one

For the empirical risk minimizer of Part I, where $M = -R$ and $M_n = -\widehat{R}_S$, stage one takes the form

$$R(\hat{\theta}_n) - \inf_{\theta \in \Theta} R(\theta) \leq 2 \sup_{\theta \in \Theta} \left| \widehat{R}_S(\theta) - R(\theta) \right| + o_{\mathbb{P}}(1) \; .$$

Names of the terms. The left side is the estimation error, Δ_n is the uniform deviation, and the $o_{\mathbb{P}}(1)$ is the optimisation error (i.e. the gap between what the optimizer returns and the true empirical minimum).

Statistical learning theory as a whole is the estimation of Δ_n, by Rademacher complexities, VC dimension or metric entropy.

Irrelevance of stage two in learning

The parameter is not identifiable, on purpose. Permuting the units of a hidden layer, or rescaling one layer against the next, leaves f_θ and hence M unchanged. As a result, the label switching counterexample of Part II is the normal situation here rather than a pathology, and $\delta \equiv 0$ below the distance between two symmetric copies.

Over-parameterized models have a continuum of exact empirical minimizers, so that $\hat{\theta}_n$ is not well defined as a point and no statement about its convergence could hold.

The model is misspecified deliberately. There is no θ_0 with $\mathcal{L}_{\theta_0} = \mathbb{P}$, and $\inf_\theta R(f_\theta) - R^\star$ is the approximation error, i.e. a deterministic quantity that the estimator cannot affect.

What is asked of a learning algorithm is the risk of the predictor it returns

Now Stage Two: Well separated maximum

Definition. A point $\theta_0 \in \Theta$ is a **well separated maximizer** of M if $M(\theta_0) = \sup_{\theta \in \Theta} M(\theta)$ and $\delta(\epsilon) > 0$ for every $\epsilon > 0$, where

$$\delta(\epsilon) := M(\theta_0) - \sup_{\theta \in \Theta : \|\theta - \theta_0\| \geq \epsilon} M(\theta) \, .$$

Reading δ. It measures, in criterion, the loss incurred by standing at distance ϵ from θ_0 in parameter, and the condition says that this loss is positive at every distance.

It is deterministic. Indeed, δ depends on M and on Θ and on nothing else (no sample, no n, no probability).

Stage two, from the criterion to the parameter

Proposition. Let θ_0 be a well separated maximizer of M, and let $(\hat{\theta}_n)$ satisfy $M(\theta_0) - M(\hat{\theta}_n) \xrightarrow{\mathbb{P}} 0$. Then $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Proof. Let us fix $\epsilon > 0$. If $\|\hat{\theta}_n - \theta_0\| \geq \epsilon$, then $\hat{\theta}_n$ belongs to the set over which the supremum defining $\delta(\epsilon)$ is taken, so that

$$\begin{aligned} M(\hat{\theta}_n) &\leq \sup_{\theta \in \Theta : \|\theta - \theta_0\| \geq \epsilon} M(\theta) \\ &= M(\theta_0) - \delta(\epsilon) \ , \end{aligned}$$

that is, $M(\theta_0) - M(\hat{\theta}_n) \geq \delta(\epsilon)$.

Hence,

$$\mathbb{P}\left(\|\hat{\theta}_n - \theta_0\| \geq \epsilon\right) \leq \mathbb{P}\left(M(\theta_0) - M(\hat{\theta}_n) \geq \delta(\epsilon)\right) \ ,$$

and the right side tends to 0 since the criterion error tends to 0 in probability and $\delta(\epsilon)$ is a fixed positive number, which concludes the proof.

The consistency theorem for M estimators

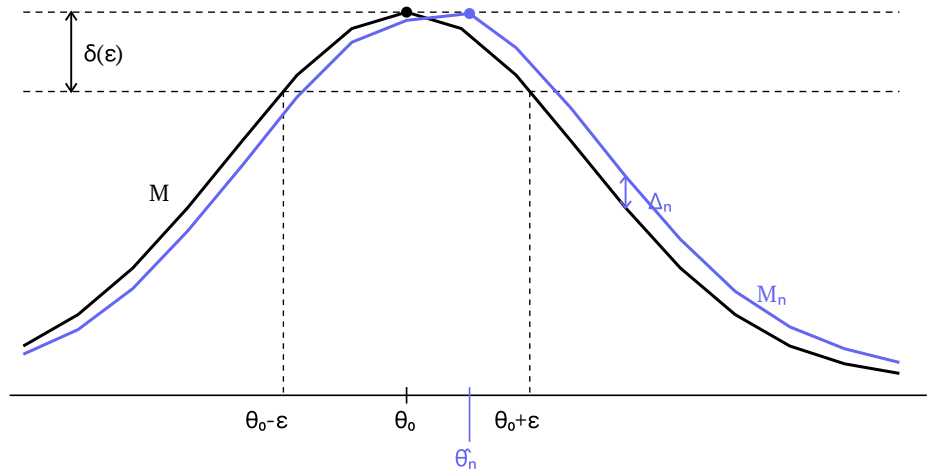
Theorem. Let (M_n) be random functions from $\Theta \subset \mathbb{R}^p$ to $\mathbb{R}$ and $M : \Theta \to \mathbb{R}$ deterministic. Assume

1. $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{\mathbb{P}} 0$,
2. there is $\theta_0 \in \Theta$ such that for every $\epsilon > 0$, $\sup_{\theta \in \Theta : \|\theta - \theta_0\| \geq \epsilon} M(\theta) < M(\theta_0)$,
3. $(\hat{\theta}_n)$ satisfies $M_n(\hat{\theta}_n) \geq \sup_{\theta \in \Theta} M_n(\theta) + o_{\mathbb{P}}(1)$.

Then $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Proof. Hypotheses 1 and 3 yield stage one and hypothesis 2 is stage two, so that composing the two propositions concludes the proof.

The two stages, in one picture



Two gaps, two stages. The blue arrow is Δ_n (i.e. how far M_n sits from M uniformly), and stage one turns it into a bound on the vertical distance from the top. The black arrow is $\delta(\epsilon)$, the exchange rate between the axes, since leaving the interval lowers the criterion by $\delta(\epsilon)$, so that a criterion error below $\delta(\epsilon)$ keeps $\hat{\theta}_n$ inside it. A flat curve makes $\delta(\epsilon)$ zero and the exchange impossible.

Role of each hypothesis

Hypothesis 1 carries all the probability. It is a law of large numbers, uniform in θ, which does not mention θ_0 and to which the final part of the lecture is entirely devoted.

Hypothesis 2 carries all the geometry. It does not mention M_n, the sample, or n.

Hypothesis 3 is the definition of the estimator, weakened as far as possible.

A sufficient condition for well separation

Lemma. If Θ is compact, M is continuous on Θ, and θ_0 is the unique maximizer of M, then θ_0 is a well separated maximizer.

Proof. Let us fix $\epsilon > 0$ and let $K_\epsilon = \{\theta \in \Theta : \|\theta - \theta_0\| \geq \epsilon\}$. If K_ϵ is empty, the supremum is $-\infty$ and there is nothing to prove. Otherwise, K_ϵ is closed in the compact Θ, hence compact, and the continuous M attains its supremum on it at some $\theta_\epsilon \neq \theta_0$, so that

$$\begin{aligned} \sup_{\theta \in K_\epsilon} M(\theta) &= M(\theta_\epsilon) \\ &< M(\theta_0)\,. \quad \text{since } \theta_0 \text{ is the unique maximizer} \end{aligned}$$

From separation to a rate

Let us suppose that the separation is polynomial, i.e. $\delta(\epsilon) \geq c\epsilon^{\alpha}$ for some $c, \alpha > 0$ and all small ϵ.

If $\|\hat{\theta}_n - \theta_0\| = r$, then $\|\hat{\theta}_n - \theta_0\| \geq \epsilon$ for every $\epsilon \leq r$, so that the argument of stage two yields $M(\theta_0) - M(\hat{\theta}_n) \geq \delta(\epsilon) \geq c\epsilon^{\alpha}$ for every $\epsilon \leq r$. Hence, $M(\theta_0) - M(\hat{\theta}_n) \geq cr^{\alpha}$ and

$$\begin{aligned} \|\hat{\theta}_n - \theta_0\| &\leq \left(\frac{M(\theta_0) - M(\hat{\theta}_n)}{c} \right)^{1/\alpha} \\ &\leq \left(\frac{2\Delta_n + o_{\mathbb{P}}(1)}{c} \right)^{1/\alpha} . \qquad \text{by stage one} \end{aligned}$$

The generic case is $\alpha = 2$. If M is twice differentiable at θ_0 with negative definite Hessian, the separation is quadratic near θ_0, and the exponent $1/\alpha = 1/2$ loses a square root, since a typical Δ_n of order $n^{-1/2}$ only yields $n^{-1/4}$.

A pessimistic rate. The bound above uses the uniform Δ_n over all of Θ to control a local quantity. We latter show that replacing it by a modulus over a shrinking neighbourhood of θ_0 recovers the rate $n^{-1/2}$.

The same two stages for Z estimators

Proposition (stage one). If $\sup_{\theta \in \Theta} \|Z_n(\theta) - Z(\theta)\| \xrightarrow{\mathbb{P}} 0$ and $Z_n(\hat{\theta}_n) = o_{\mathbb{P}}(1)$, then $\|Z(\hat{\theta}_n)\| = o_{\mathbb{P}}(1)$.

Proof. By the triangle inequality, $\|Z(\hat{\theta}_n)\| \leq \|Z_n(\hat{\theta}_n)\| + \sup_{\theta \in \Theta} \|Z_n(\theta) - Z(\theta)\|$, and both terms are $o_{\mathbb{P}}(1)$.

Proposition (stage two). If $Z(\theta_0) = 0$ and $\delta(\epsilon) := \inf_{\|\theta - \theta_0\| \geq \epsilon} \|Z(\theta)\| > 0$ for every $\epsilon > 0$, then $\|Z(\hat{\theta}_n)\| = o_{\mathbb{P}}(1)$ implies $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Proof. $\|\hat{\theta}_n - \theta_0\| \geq \epsilon$ forces $\|Z(\hat{\theta}_n)\| \geq \delta(\epsilon)$, so $\mathbb{P}(\|\hat{\theta}_n - \theta_0\| \geq \epsilon) \leq \mathbb{P}(\|Z(\hat{\theta}_n)\| \geq \delta(\epsilon)) \to 0$.

The consistency theorem for Z estimators

Theorem. Let (Z_n) be random functions from $\Theta \subset \mathbb{R}^p$ to $\mathbb{R}^q$ and $Z : \Theta \to \mathbb{R}^q$ deterministic. Assume

1. $\sup_{\theta \in \Theta} \|Z_n(\theta) - Z(\theta)\| \xrightarrow{\mathbb{P}} 0$,
2. $Z(\theta_0) = 0$ and, for every $\epsilon > 0$, $\inf_{\theta \in \Theta : \|\theta - \theta_0\| \geq \epsilon} \|Z(\theta)\| > 0$,
3. $(\hat{\theta}_n)$ satisfies $Z_n(\hat{\theta}_n) = o_{\mathbb{P}}(1)$.

Then $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

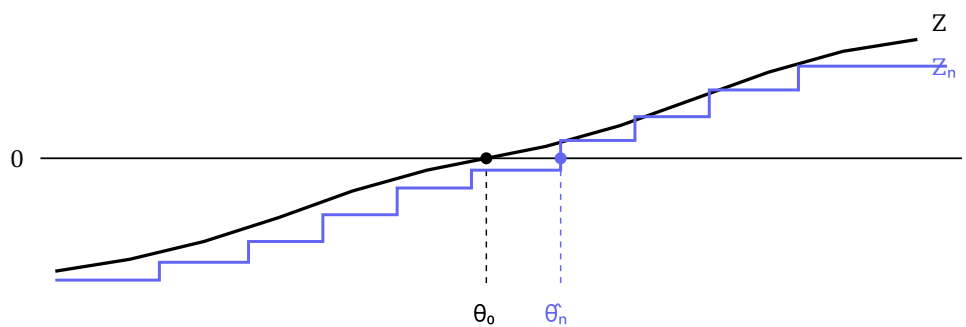
Remark. The course notes write hypothesis 2 as one chain, $\inf \|Z(\theta)\| > 0 = \|Z(\theta_0)\|$. It consists of two separate requirements, namely that the limit function vanishes at θ_0 and that it stays away from zero everywhere else.

Easy case, a monotone function in one dimension

Theorem. Let $\Theta = \mathbb{R}$, let (Z_n) be random functions from $\mathbb{R}$ to $\mathbb{R}$ and $Z : \mathbb{R} \to \mathbb{R}$ deterministic. Assume

1. for every **fixed** θ, $Z_n(\theta) \xrightarrow{\mathbb{P}} Z(\theta)$, with no uniformity,
2. almost surely, Z_n is non-decreasing,
3. there is θ_0 such that $Z(\theta_0 - \epsilon) < 0 < Z(\theta_0 + \epsilon)$ for every $\epsilon > 0$.

Then any $(\hat{\theta}_n)$ with $Z_n(\hat{\theta}_n) = o_{\mathbb{P}}(1)$ satisfies $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.



Proof in the monotone case

Let us fix $\epsilon > 0$. Splitting on the two sides, we have

$$
\begin{aligned}
&\mathbb{P}\left(|\hat{\theta}_n - \theta_0| \geq \epsilon\right) \\
&= \mathbb{P}(\hat{\theta}_n \leq \theta_0 - \epsilon) + \mathbb{P}(\hat{\theta}_n \geq \theta_0 + \epsilon) \\
&\leq \mathbb{P}\left(Z_n(\hat{\theta}_n) \leq Z_n(\theta_0 - \epsilon)\right) + \mathbb{P}\left(Z_n(\hat{\theta}_n) \geq Z_n(\theta_0 + \epsilon)\right) && \text{since } Z_n \text{ is non-decreasing} \\
&= \mathbb{P}\left(o_{\mathbb{P}}(1) \leq Z(\theta_0 - \epsilon) + o_{\mathbb{P}}(1)\right) + \mathbb{P}\left(o_{\mathbb{P}}(1) \geq Z(\theta_0 + \epsilon) + o_{\mathbb{P}}(1)\right) \ . && \text{by hypotheses 3 and 1}
\end{aligned}
$$

The two numbers $Z(\theta_0 - \epsilon) < 0$ and $Z(\theta_0 + \epsilon) > 0$ are fixed, and each probability is the probability that a quantity tending to 0 in probability exceeds a fixed nonzero threshold in the wrong direction, so that both tend to 0, which concludes the proof.

The trade-off. Uniform convergence, the demanding hypothesis of the general theorem, is replaced by pointwise convergence at two points per ϵ. In exchange, the statement needs monotonicity and dimension one.

Exercise, the empirical median

Exercise. Let $X_1, X_2, \ldots$ be i.i.d. copies of a real X with a continuous density bounded away from 0 on a neighbourhood of the population median. Show that the empirical median $\hat{\theta}_n$, defined by $\sum_{i \le n} \operatorname{sign}(\hat{\theta}_n - X_i) = 0$, is consistent.

Solution. We take $Z_n(\theta) = \frac{1}{n} \sum_{i \le n} \operatorname{sign}(\theta - X_i)$, so that $Z_n(\hat{\theta}_n) = 0$, and

$$\begin{aligned} Z_n(\theta) &= \frac{1}{n} \sum_{i \le n} \mathbf{1}\{X_i < \theta\} - \frac{1}{n} \sum_{i \le n} \mathbf{1}\{X_i > \theta\} \\ &\xrightarrow{\mathbb{P}} \mathbb{P}(X < \theta) - \mathbb{P}(X > \theta) && \text{by the law of large numbers} \\ &= 2F_X(\theta) - 1\,, && \text{since } \mathbb{P}(X = \theta) = 0 \end{aligned}$$

which we denote $Z(\theta)$.

Hypothesis 2 holds since $\theta \mapsto \operatorname{sign}(\theta - x)$ is non-decreasing for each x, and hypothesis 3 holds with θ_0 the median ($F_X(\theta_0) = 1/2$), since F_X is strictly increasing where the density is positive. Hence, the theorem applies.

Part III. Uniform convergence

The uniform deviation

For both consistency theorems to apply, we need the uniform deviation of stage one to tend to 0,

$$\Delta_n = \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{\mathbb{P}} 0 \ .$$

Simplification. Let $X_1, X_2, \ldots$ i.i.d... We consider $M_n(\theta) = \frac{1}{n} \sum_{i \leq n} m(X_i, \theta)$ and $M(\theta) = \mathbb{E}_X[m(X, \theta)]$ for a function $m : \mathbb{R}^d \times \Theta \to \mathbb{R}$.

Pointwise VS Uniform convergence. For each θ, the law of large numbers yields $M_n(\theta) \xrightarrow{\mathbb{P}} M(\theta)$, but uniformity in θ is out of its reach.

Rewriting. With $\mathcal{F} = \{m(\cdot, \theta); \theta \in \Theta\}$, the uniform deviation becomes

$$\Delta_n = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i \leq n} f(X_i) - \mathbb{E}_X[f(X)] \right| \ .$$

Glivenko-Cantelli classes. With $\mathcal{L}$ the law of X, we say that $\mathcal{F}$ is **$\mathcal{L}$-Glivenko-Cantelli** if $\Delta_n \xrightarrow{\mathbb{P}} 0$.

First idea: Covering in supremum norm

If $\mathcal{F}$ is finite, the law of large numbers applied to each element and a maximum over finitely many terms ssolves the problem by union bound.

Otherwise, an idea could be to approximate $\mathcal{F}$ by a finite family of functions. If every $f \in \mathcal{F}$ has some f' in a finite $\mathcal{F}'$ with $\|f - f'\|_\infty \leq \epsilon$, then

$$\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i \leq n} f(X_i) - \mathbb{E}_X[f(X)] \right| \leq \sup_{f' \in \mathcal{F}'} \left| \frac{1}{n} \sum_{i \leq n} f'(X_i) - \mathbb{E}_X[f'(X)] \right| + 2\epsilon \, .$$

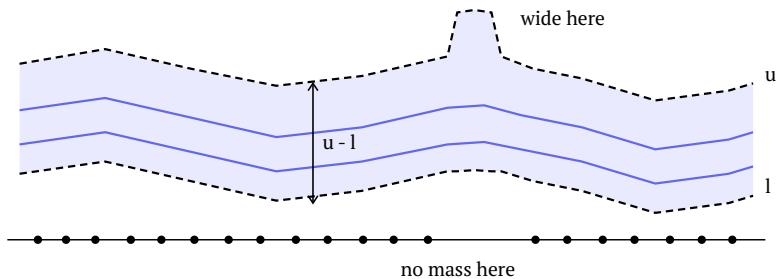
Exercise. Compute the smallest cardinality of such an $\mathcal{F}'$ for $\mathcal{F} = \{\mathbf{1}_{(-\infty,t]}; t \in [0,1]\}$.

Solution. For $s \neq t$, the two indicators differ by 1 somewhere, so that $\|\mathbf{1}_{(-\infty,t]} - \mathbf{1}_{(-\infty,s]}\|_\infty = 1$. Hence, for $\epsilon < 1/2$, a ball of radius ϵ contains at most one element of $\mathcal{F}$, and since $\mathcal{F}$ is uncountable, no finite $\mathcal{F}'$ works. The covering number is infinite on the simplest class there is, which is nevertheless $\mathcal{L}$-Glivenko-Cantelli for every law $\mathcal{L}$ by the classical Glivenko-Cantelli theorem.

Refinement: Brackets

Definition. For $l \le u$ pointwise, the **bracket** $[l, u]$ is the set of functions $f : \mathbb{R}^d \to \mathbb{R}$ with $l \le f \le u$ pointwise. For $\epsilon > 0$, $q > 0$ and a law $\mathcal{L}$ on $\mathbb{R}^d$, the **bracketing number** $N_{[]}(\mathcal{F}, L^q(\mathcal{L}), \epsilon)$ is the smallest N for which some brackets $[l_1, u_1], \ldots, [l_N, u_N]$ cover $\mathcal{F}$ and satisfy, for every j,

$$\left(\int (u_j - l_j)^q d\mathcal{L} \right)^{1/q} \le \epsilon \, .$$

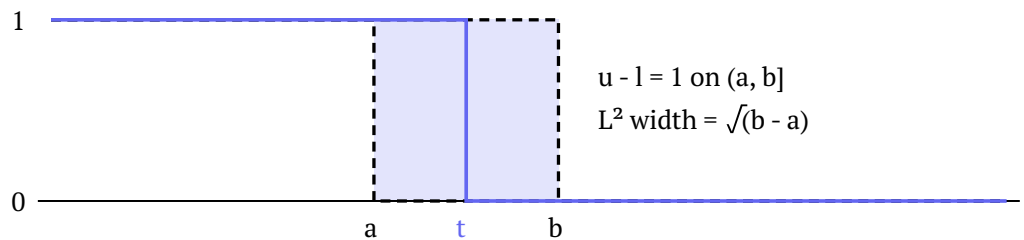


Two relaxations. The width is measured in $L^q(\mathcal{L})$ rather than in supremum norm, and l, u (drawn dashed) need not belong to $\mathcal{F}$.

Exercise, the indicator class

Exercise. Let $\mathcal{L} = \mathrm{Unif}([0,1])$ and $\mathcal{F} = \{f_t = \mathbf{1}_{(-\infty,t]}; t \in [0,1]\}$. Find $N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon)$ up to multiplicative constants.

Solution, upper bound. For $s \leq t$ in $[0,1]$, $\|f_t - f_s\|_{L^2(\mathcal{L})} = \sqrt{t-s}$, and $f_r \in [f_s, f_t]$ for every $r \in [s,t]$. Hence, the brackets $[f_0, f_{\epsilon^2}], [f_{\epsilon^2}, f_{2\epsilon^2}], \ldots$ cover $\mathcal{F}$, have width at most ϵ, and there are $\lceil \epsilon^{-2} \rceil$ of them.



Solution, lower bound. If $f_s, f_t \in [l,u]$, then $u - l \geq |f_t - f_s|$ pointwise, so $\|u-l\|_{L^2(\mathcal{L})} \geq \sqrt{|t-s|}$. A bracket of width ϵ thus contains f_t only for t in a set of diameter at most ϵ^2, and covering $[0,1]$ requires at least ϵ^{-2} of them. Hence, $N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon) \asymp \epsilon^{-2}$.

Consistency theorem: Finite bracketing complexity => Glivenko-Cantelli

Theorem. Let $\mathcal{F}$ be a class of $\mathcal{L}$-integrable functions with $N_{[]}(\mathcal{F}, L^1(\mathcal{L}), \epsilon) < \infty$ for every $\epsilon > 0$. Then $\mathcal{F}$ is $\mathcal{L}$-Glivenko-Cantelli.

Proof. Let us fix $\epsilon > 0$, let $N = N_{[]}(\mathcal{F}, L^1(\mathcal{L}), \epsilon)$ and let $[l_j, u_j], j \leq N$, be brackets covering $\mathcal{F}$ with $\int |u_j - l_j| d\mathcal{L} \leq \epsilon$. Discarding the brackets that meet no element of $\mathcal{F}$, each l_j and u_j is within ϵ in L^1 of an integrable function, hence integrable.

Given $f \in \mathcal{F}$, we choose j with $l_j \leq f \leq u_j$. Then, $\mathbb{E}_X[l_j(X)] \leq \mathbb{E}_X[f(X)] \leq \mathbb{E}_X[u_j(X)] \leq \mathbb{E}_X[l_j(X)] + \epsilon$, and averaging the same inequalities over the sample,

$$\frac{1}{n}\sum_{i\leq n} l_j(X_i) \leq \frac{1}{n}\sum_{i\leq n} f(X_i) \leq \frac{1}{n}\sum_{i\leq n} u_j(X_i)\ .$$

Proof, continued

Subtracting $\mathbb{E}_X[f(X)]$ and using the two displayed chains, we obtain

$$\frac{1}{n}\sum_{i\leq n} l_j(X_i) - \mathbb{E}_X[l_j(X)] - \epsilon \leq \frac{1}{n}\sum_{i\leq n} f(X_i) - \mathbb{E}_X[f(X)] \leq \frac{1}{n}\sum_{i\leq n} u_j(X_i) - \mathbb{E}_X[u_j(X)] + \epsilon\,.$$

Since the bound only depends on f through j, taking the supremum over $\mathcal{F}$ yields

$$\sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i\leq n} f(X_i) - \mathbb{E}_X[f(X)] \right| \leq \max_{g\in\{l_1,u_1,\ldots,l_N,u_N\}} \left| \frac{1}{n}\sum_{i\leq n} g(X_i) - \mathbb{E}_X[g(X)] \right| + \epsilon\,.$$

The maximum is over $2N$ fixed integrable functions. By the law of large numbers, each term tends to 0 almost surely, and a maximum of finitely many sequences tending to 0 tends to 0. Hence, $\limsup_n \mathbb{P}(\sup_{\mathcal{F}} |\cdot| \geq 2\epsilon) = 0$ for every $\epsilon > 0$, which concludes the proof.

Corollary: Continuously parametrized functions over metric spaces.

Proposition. Let $\mathcal{F} = \{g_\theta; \theta \in \Theta\}$ with $g_\theta : \mathbb{R}^d \to \mathbb{R}$, and assume

1. Θ is a compact metric space,
2. $\theta \mapsto g_\theta(x)$ is continuous for every x,
3. $\int \sup_{\theta \in \Theta} |g_\theta(x)| d\mathcal{L}(x) < \infty$.

Then $\mathcal{F}$ is $\mathcal{L}$-Glivenko-Cantelli.

The three hypotheses are those that a model satisfies in practice, namely a compact parameter set, a criterion continuous in the parameter, and one integrable envelope.

Proof, building the brackets

Let us fix $\epsilon > 0$. For $\theta \in \Theta$ and $N \in \mathbb{N}$, let $B_{\theta,N}$ be the open ball of radius $1/N$ around θ in Θ, and

$$\tilde{l}_{\theta,N}(x) = \inf_{\tilde{\theta} \in B_{\theta,N}} g_{\tilde{\theta}}(x) \ ,$$
$$\tilde{u}_{\theta,N}(x) = \sup_{\tilde{\theta} \in B_{\theta,N}} g_{\tilde{\theta}}(x) \ .$$

For fixed x, $\tilde{u}_{\theta,N}(x) - \tilde{l}_{\theta,N}(x) \to 0$ as $N \to \infty$ by continuity of $\theta \mapsto g_\theta(x)$ at θ. Since the difference is dominated by $2 \sup_{\theta \in \Theta} |g_\theta|$, which is integrable by hypothesis 3, dominated convergence yields

$$\int \left| \tilde{u}_{\theta,N} - \tilde{l}_{\theta,N} \right| d\mathcal{L} \xrightarrow[N \to \infty]{} 0 \ .$$

We choose N_θ with $\int |\tilde{u}_{\theta,N_\theta} - \tilde{l}_{\theta,N_\theta}| d\mathcal{L} \leq \epsilon$, and fix it.

Proof, a finite subcover

The balls $B_{\theta,N_\theta}, \theta \in \Theta$, form an open cover of Θ, and by compactness, finitely many of them suffice:

$$\Theta \subset \bigcup_{j \leq m} B_{\theta_j, N_{\theta_j}} \ .$$

We set $l_j = \tilde{l}_{\theta_j,N_{\theta_j}}$ and $u_j = \tilde{u}_{\theta_j,N_{\theta_j}}$. Every $\theta \in \Theta$ lies in some $B_{\theta_j,N_{\theta_j}}$, and then $l_j \leq g_\theta \leq u_j$ by construction, so that the m brackets $[l_j, u_j]$ cover $\mathcal{F}$ and each has $L^1(\mathcal{L})$ width at most ϵ.

Hence $N_{[]}(\mathcal{F}, L^1(\mathcal{L}), \epsilon) \leq m < \infty$ for every $\epsilon > 0$, and an application of the previous theorem concludes the proof.

Role of compactness. It only appears here, to pass from a cover indexed by Θ to a finite subcover, which is the role it played in the well separation lemma of Part II, and the reason why both appear in the maximum likelihood theorem.

Consistency of maximum likelihood

We now apply Parts II and III to our running example, and prove that maximum likelihood is consistent.

Theorem. Let $\{\mathcal{L}_\theta; \theta \in \Theta\}$ have densities f_θ with respect to Lebesgue measure, and let $X_1, X_2, \ldots$ be i.i.d. with density f_{θ_0}. Assume

1. Θ is compact in $\mathbb{R}^p$,
2. $f_\theta(x) > 0$ for every θ and x,
3. $\theta \mapsto f_\theta(x)$ is continuous on Θ for every x,
4. $\int \sup_{\theta \in \Theta} |\log f_\theta(x)| f_{\theta_0}(x) dx < \infty$,
5. **identifiability**: $\mathcal{L}_\theta \neq \mathcal{L}_{\theta_0}$ for every $\theta \neq \theta_0$.

Then the maximum likelihood estimator satisfies $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Plan of the proof

Maximum likelihood is the M-estimator with $M_n(\theta) = \frac{1}{n}\sum_{i\leq n} \log f_\theta(X_i)$, whose pointwise limit is $M(\theta) = \mathbb{E}_X[\log f_\theta(X)]$ by the law of large numbers. We apply the consistency theorem for M estimators, whose three hypotheses are the three steps of the proof.

1. **Uniform convergence**, i.e. $\{\log f_\theta; \theta \in \Theta\}$ is $\mathcal{L}_{\theta_0}$-Glivenko-Cantelli. This is where Part III comes into play, since items 1 to 4 yield the hypotheses of the corollary on continuously parametrized functions.

2. **Well separation of θ_0.** Identifiability (item 5) makes θ_0 the unique maximizer of M, and the lemma of Part II turns it into well separation, with compactness (item 1) and the continuity of M.

3. **Near maximization**, which holds by definition of $\hat{\theta}_n$.

Proof, uniform convergence

For the first step, we apply the corollary on continuously parametrized functions with $g_\theta = \log f_\theta$. Its hypothesis 1 is item 1, its hypothesis 2 follows from items 2 and 3, and its hypothesis 3 is item 4, since $d\mathcal{L}_{\theta_0}(x) = f_{\theta_0}(x)dx$. Hence, $\{\log f_\theta; \theta \in \Theta\}$ is $\mathcal{L}_{\theta_0}$-Glivenko-Cantelli, which means that

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i \leq n} \log f_\theta(X_i) - \mathbb{E}_X[\log f_\theta(X)] \right| \xrightarrow{\mathbb{P}} 0 \ .$$

We thus obtain hypothesis 1 of the consistency theorem for M estimators, which concludes the first step.

Proof, the separation

For the second step, we first show that identifiability makes θ_0 the unique maximizer of M, using the inequality $\log t \leq 2(\sqrt{t} - 1)$ for $t > 0$. For $\theta \neq \theta_0$, we have

$$
\begin{aligned}
M(\theta) - M(\theta_0) &= \int \log\left(\frac{f_\theta(x)}{f_{\theta_0}(x)}\right) f_{\theta_0}(x)dx && \text{by definition of } M \\
&\leq 2\int \left(\sqrt{\frac{f_\theta(x)}{f_{\theta_0}(x)}} - 1\right) f_{\theta_0}(x)dx && \text{by the inequality above} \\
&= -\int \left(\sqrt{f_\theta(x)} - \sqrt{f_{\theta_0}(x)}\right)^2 dx\,, && \text{since } \textstyle\int f_\theta = \int f_{\theta_0} = 1
\end{aligned}
$$

which is negative by item 5. Hence, θ_0 is the unique maximizer of M.

M is continuous on Θ by dominated convergence (items 2 and 3 yield the continuity of the integrand and item 4 the domination). With item 1, the lemma of Part II shows that θ_0 is a well separated maximizer. Since the third step holds by definition, the consistency theorem concludes the proof.

Relaxing identifiability: The risk gap of maximum likelihood

Without identifiability, the first step alone still says something about the estimated law $\mathcal{L}_{\hat{\theta}_n}$. Since maximum likelihood is the empirical risk minimizer for the log loss, stage one controls its risk gap, which we now identify.

Proposition. If the model is well specified, then, with $R(\theta) = \mathbb{E}_X[-\log f_\theta(X)]$, the risk gap of the log loss is a relative entropy:

$$R(\theta) - \inf_{\theta' \in \Theta} R(\theta') = \mathrm{KL}(\mathcal{L}_{\theta_0} \| \mathcal{L}_\theta) \ .$$

Proof. We have

$$\begin{aligned} R(\theta) - R(\theta_0) &= \mathbb{E}_X \left[\log \frac{f_{\theta_0}(X)}{f_\theta(X)} \right] \\ &= \mathrm{KL}(\mathcal{L}_{\theta_0} \| \mathcal{L}_\theta) \\ &\geq 0 \ , \qquad \text{by Jensen's inequality} \end{aligned}$$

Consistency of the estimated law

Plugging this gap into empirical risk minimization as stage one, with Δ_n the uniform deviation of the class $\{\log f_\theta ; \theta \in \Theta\}$, we obtain

$$\mathrm{KL}(\mathcal{L}_{\theta_0} \| \mathcal{L}_{\hat{\theta}_n}) \leq 2\Delta_n + o_{\mathbb{P}}(1) \ .$$

The estimated law is consistent in relative entropy as soon as this class is $\mathcal{L}_{\theta_0}$-Glivenko-Cantelli, without identifiability, compactness or separation. In particular, fitting a non-identifiable model still estimates something, since the predicted distribution converges and only the parameter is meaningless.

Role of identifiability. It only serves to pass from the law to the parameter, in the second step of the proof.

Part IV. Limit laws

Beyond consistency

Parts II and III prove that $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$. However, consistency only says where the estimator goes, and nothing about the size of the error $\hat{\theta}_n - \theta_0$ at a given n.

The empirical mean as a guide. For $\bar{X}_n$, consistency is the law of large numbers, and the central limit theorem goes one step further. For real X with mean μ and variance σ^2, it states that

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2) \ .$$

Hence, the error is of order $\sigma/\sqrt{n}$, and $[\bar{X}_n \pm 1.96\sigma/\sqrt{n}]$ contains μ with probability going to 0.95.

Goal of this part. We look for the analogue of the central limit theorem for $\hat{\theta}_n$, called **asymptotic normality**, i.e. for a covariance matrix Σ such that

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Sigma) \ .$$

The heuristic, linearization

Unlike $\bar{X}_n$, the estimator $\hat{\theta}_n$ is not an average. For the method of moments of the previous lecture, it was an explicit function $e^{-1}(e_n)$ of an average, and the delta method concluded. In general, $\hat{\theta}_n$ is only defined implicitly by $Z_n(\hat{\theta}_n) = 0$, and the idea is to linearize Z_n around θ_0 to come back to averages.

Let us take $Z_n(\theta) = \frac{1}{n}\sum_{i \leq n} z(X_i, \theta)$ with X_i i.i.d. and $\mathbb{E}_X[z(X, \theta_0)] = 0$. Since $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$ by Parts II and III, we have

$$\begin{aligned} 0 &= Z_n(\hat{\theta}_n) && \text{by definition of } \hat{\theta}_n \\ &\approx Z_n(\theta_0) + (JZ_n)(\theta_0)(\hat{\theta}_n - \theta_0)\ . && \text{by a first order expansion at } \theta_0 \end{aligned}$$

Solving for the error and multiplying by $\sqrt{n}$, we obtain

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \approx -((JZ_n)(\theta_0))^{-1}\sqrt{n}Z_n(\theta_0)\ ,$$

where $(JZ_n)(\theta_0)$ and $Z_n(\theta_0)$ are both averages of i.i.d. terms.

The heuristic, the limit law

Each of the two averages is handled by a classical theorem.

Law of large numbers. $(JZ_n)(\theta_0) \xrightarrow{\mathbb{P}} \mathbb{E}_X[J_z(X, \theta_0)] =: V$, and if V is invertible, continuous mapping inverts it.

Central limit theorem. $\sqrt{n} Z_n(\theta_0) \xrightarrow{d} \mathcal{N}(0, \mathrm{Cov}_X(z(X, \theta_0)))$, where no centering is needed since $\mathbb{E}_X[z(X, \theta_0)] = 0$.

Slutsky. Combining the two, we obtain asymptotic normality,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, V^{-1} \mathrm{Cov}_X(z(X, \theta_0)) V^{-\top}\right) \ .$$

Limits of the heuristic

Warning. The expansion requires $\theta \mapsto z(x, \theta)$ to be differentiable for each fixed x. However, for the empirical median, $z(x, \theta) = \operatorname{sign}(\theta - x)$ is not differentiable at $\theta = x$, and not even continuous there, so that the main example of this lecture is out of the reach of the heuristic.

What survives. Only $\theta \mapsto \mathbb{E}_X[z(X, \theta)]$ needs to be smooth. For the median, this expectation is $2F_X(\theta) - 1$, which is as smooth as the density. In other words, averaging over x repairs what happens at each fixed x.

What replaces the remainder control. The heuristic implicitly controls a Taylor remainder at the random point $\hat{\theta}_n$. We replace it by a uniform bound over a shrinking neighbourhood of θ_0, obtained with the maximal inequality of the next slide, and the same bound yields the limit law of the value.

The main tool: the maximal inequality

Theorem. Let $X_1, X_2, \ldots$ be i.i.d. of law $\mathcal{L}$ on $\mathbb{R}^d$ and let $\mathcal{F}$ be a class of functions $\mathbb{R}^d \to \mathbb{R}$ with an **envelope** F, meaning $|f(x)| \leq F(x)$ for every $x \in \mathbb{R}^d$ and every $f \in \mathcal{F}$, and $\mathbb{E}_X[F(X)^2] < \infty$. Then

$$\mathbb{E}^\star\left[\sup_{f\in\mathcal{F}}\left(\frac{1}{n}\sum_{i\leq n} f(X_i) - \mathbb{E}_X[f(X)]\right)\right] \leq \frac{C_{\mathrm{MI}}}{\sqrt{n}}\int_0^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{\log N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon)}\, d\epsilon\ ,$$

for a universal constant C_{MI}.

We omit the proof, which can be found in van der Vaart, Corollary 19.35, and we sketch it below. Here, $\mathbb{E}^\star$ is the outer expectation, since a supremum over an uncountable class need not be measurable.

The empirical process. Denoting $\mathbb{G}_n f = \frac{1}{\sqrt{n}}\sum_{i\leq n}(f(X_i) - \mathbb{E}_X[f(X)])$, the theorem equivalently states that $\mathbb{E}^\star[\sup_{f\in\mathcal{F}} \mathbb{G}_n f]$ is at most C_{MI} times the integral, which is the form used in the proof.

Absolute values add a 1. We have $\sup_f |\mathbb{G}_n f| = \sup_{g\in\mathcal{F}\cup(-\mathcal{F})} \mathbb{G}_n g$, and negating a bracket keeps its width, so bracketing numbers at most double. Hence, the theorem on $\mathcal{F}\cup(-\mathcal{F})$ yields the same bound with $\sqrt{1+\log N_{[]}}$, as $\log 2 \leq 1$. **We use this form below.**

Sketch of the proof

A single resolution. We call **resolution** ϵ a finite family of brackets covering $\mathcal{F}$, of width at most ϵ in a norm that we choose below. For a bracket $l \leq f \leq u$ of this family, the Glivenko-Cantelli argument of Part III yields

$$\frac{1}{n}\sum_{i\leq n} f(X_i) - \mathbb{E}_X[f(X)] \leq \underbrace{\frac{1}{n}\sum_{i\leq n} u(X_i) - \mathbb{E}_X[u(X)]}_{\text{fluctuation}} + \underbrace{\|u-l\|_{L^1(\mathcal{L})}}_{\text{bias}} .$$

Without $\sqrt{n}$, in Part III. The fluctuation tends to 0 by the law of large numbers, so that only the bias remains, and widths measured in L^1 were enough.

With $\sqrt{n}$, here. The fluctuation becomes $\mathbb{G}_n u$, of order $\|u\|_{L^2(\mathcal{L})}$, and the bias becomes $\sqrt{n}\|u-l\|_{L^1(\mathcal{L})}$, which forces widths $\lesssim n^{-1/2}$ and far too many brackets for a union bound. Since the fluctuations are of L^2 size, and since $\|u-l\|_{L^1(\mathcal{L})} \leq \|u-l\|_{L^2(\mathcal{L})}$ by Cauchy-Schwarz, we measure widths in L^2 from now on.

Chaining. We approximate f at every resolution $\epsilon_k = 2^{-k}\epsilon_0$ at once and telescope, $f = f_0 + \sum_{k\geq 1}(f_k - f_{k-1})$. Only fluctuations of the increments then remain, and no bias of order $\sqrt{n}\epsilon$ appears.

Bernstein's inequality

Each increment $h = f_k - f_{k-1}$ appears through the centered average $\mathbb{G}_n h$, whose tail we control as follows.

Lemma (Bernstein's inequality). Let h be bounded with $\mathbb{E}_X[h(X)] = 0$. Then, for every $t > 0$,

$$\mathbb{P}\left(|\mathbb{G}_n h| \geq t\right) \leq 2 \exp\left(-\frac{t^2/2}{\|h\|^2_{L^2(\mathcal{L})} + t\|h\|_\infty/(3\sqrt{n})}\right).$$

We omit the proof, which relies on the Chernoff method (Boucheron, Lugosi and Massart, Section 2.8).

Variance term. $\|h\|^2_{L^2(\mathcal{L})} = \mathrm{Var}_X(h(X))$ is the variance of $\mathbb{G}_n h$ and of its gaussian limit.

Range term. $t\|h\|_\infty/(3\sqrt{n})$ accounts for the largest value of h, and vanishes as $n \to \infty$.

Two regimes. If $t \leq 3\sqrt{n}\|h\|^2_{L^2(\mathcal{L})}/\|h\|_\infty$, the variance term dominates and the bound is at most the gaussian tail $2\exp(-t^2/(4\|h\|^2_{L^2(\mathcal{L})}))$. Beyond, the range term dominates and the tail is only exponential.

Origin of the L^2 control

Since the threshold of the gaussian regime grows like $\sqrt{n}$, Bernstein's inequality makes $\mathbb{G}_n h$ approximately sub-gaussian for large n, with variance proxy $\|h\|^2_{L^2(\mathcal{L})}$. Hence, a maximum over N such increments satisfies

$$\mathbb{E}_S\left[\max_{j\le N} \mathbb{G}_n h_j\right] \lesssim \max_{j\le N} \|h_j\|_{L^2(\mathcal{L})}\sqrt{\log N}\ .$$

Reading the bound. A maximum of N increments grows like their largest standard deviation times $\sqrt{\log N}$, and the standard deviation of $\mathbb{G}_n h$ is $\|h\|_{L^2(\mathcal{L})}$. This is where the L^2 norm comes from.

L^1 against L^2. The L^1 norm of Part III measured the bias of a bracket, which chaining cancels, whereas the L^2 norm measures the fluctuations of the increments, which chaining keeps.

Construction of the chain

Let $\epsilon_0 = 2\sqrt{\mathbb{E}_X[F(X)^2]}$, the L^2 width of the bracket $[-F, F]$, and, for $k \geq 0$, let $\epsilon_k = 2^{-k}\epsilon_0$ and $N_k = N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon_k)$, so that $N_0 = 1$.

The approximation f_k. We fix N_k brackets of L^2 width at most ϵ_k covering $\mathcal{F}$, and we pick one element of $\mathcal{F}$ in each of them. To $f \in \mathcal{F}$, we associate the element f_k picked in a bracket $[l, u]$ containing f. Hence, f_k takes at most N_k values as f varies, and since $|f - f_k| \leq u - l$, we have $\|f - f_k\|_{L^2(\mathcal{L})} \leq \epsilon_k$. In particular, f_0 is the same for every f.

The increments. The increment $f_k - f_{k-1}$ only depends on the brackets of f at resolutions $k-1$ and k, so that it belongs to a set $\mathcal{H}_k$ of at most $N_{k-1}N_k \leq N_k^2$ functions. Furthermore, we have

$$\begin{aligned}
\|f_k - f_{k-1}\|_{L^2(\mathcal{L})} &\leq \|f_k - f\|_{L^2(\mathcal{L})} + \|f - f_{k-1}\|_{L^2(\mathcal{L})} && \text{by the triangle inequality} \\
&\leq \epsilon_k + \epsilon_{k-1} \\
&= 3\epsilon_k \ . && \text{since } \epsilon_{k-1} = 2\epsilon_k
\end{aligned}$$

The entropy integral, resolution by resolution

Telescoping the chain, the supremum over f is at most $\mathbb{G}_n f_0$ plus, at each resolution k, a maximum over the finite set $\mathcal{H}_k$. Since f_0 is fixed, $\mathbb{E}_S[\mathbb{G}_n f_0] = 0$, and we have

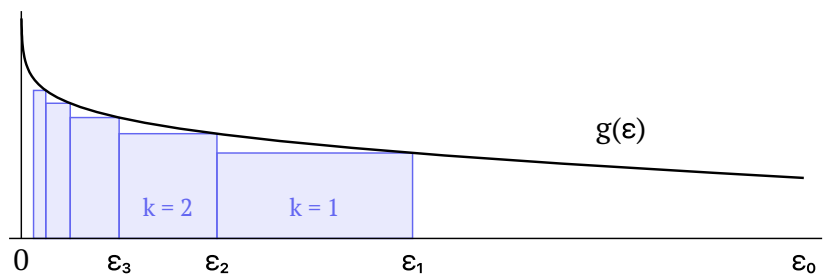
$$\begin{aligned}
\mathbb{E}^\star\left[\sup_{f\in\mathcal{F}} \mathbb{G}_n f\right] &\le \sum_{k\ge 1} \mathbb{E}_S\left[\max_{h\in\mathcal{H}_k} \mathbb{G}_n h\right] \\
&\lesssim \sum_{k\ge 1} 3\epsilon_k \sqrt{\log(N_k^2)} \qquad \text{by the bound on a maximum of increments} \\
&= 3\sqrt{2}\sum_{k\ge 1} \epsilon_k \sqrt{\log N_k}\ .
\end{aligned}$$

Reading the sum. Each resolution contributes the size ϵ_k of its increments times the square root of the logarithm of their number. It remains to compare this sum with the entropy integral.

The entropy integral as a Riemann sum

We denote $g(\epsilon) = \sqrt{\log N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon)}$, which is non-increasing in ϵ. Since $\epsilon_k - \epsilon_{k+1} = \epsilon_k/2$, the sum over resolutions is twice a Riemann sum of g, evaluated at the right end of each interval, where g is smallest. Hence, we have

$$\begin{aligned}
\sum_{k\geq 1} \epsilon_k g(\epsilon_k) &= 2 \sum_{k\geq 1} (\epsilon_k - \epsilon_{k+1}) g(\epsilon_k) && \text{since } \epsilon_{k+1} = \epsilon_k/2 \\
&\leq 2 \sum_{k\geq 1} \int_{\epsilon_{k+1}}^{\epsilon_k} g(\epsilon) d\epsilon && \text{since } g \geq g(\epsilon_k) \text{ on } [\epsilon_{k+1}, \epsilon_k] \\
&\leq 2 \int_0^{\epsilon_0} g(\epsilon) d\epsilon \ . && \text{since these intervals tile } (0, \epsilon_1]
\end{aligned}$$



rectangle k, of width $\varepsilon_k/2$ and height $g(\varepsilon_k)$, lies under the graph

Exercise, the empirical distribution function

Exercise. Let $X_1, X_2, \ldots$ be i.i.d. $\mathrm{Unif}([0,1])$ and $\mathcal{F} = \{\mathbf{1}_{(-\infty,t]}; t \in [0,1]\}$. Bound $\mathbb{E}^\star[\sup_{f\in\mathcal{F}} \frac{1}{\sqrt{n}} |\sum_{i\le n}(f(X_i) - \mathbb{E}_X[f(X)])|]$.

Solution. The constant function $F \equiv 1$ is an envelope, so that $\mathbb{E}_X[F(X)^2] = 1$, and the exercise of Part III yields $N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon) \lesssim \epsilon^{-2}$, hence

$$
\begin{aligned}
&\int_0^1 \sqrt{1 + \log N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon)}\, d\epsilon \\
&\lesssim \int_0^1 \left(1 + \sqrt{2\log(1/\epsilon)}\right) d\epsilon && \text{since } \sqrt{1+x} \le 1 + \sqrt{x} \\
&= 1 + \sqrt{2}\int_0^\infty \sqrt{u} e^{-u} du && \text{with } \epsilon = e^{-u} \\
&= 1 + \frac{\sqrt{2\pi}}{2}\,. && \text{since } \sqrt{2}\Gamma(3/2) = \sqrt{2\pi}/2
\end{aligned}
$$

As a result, the empirical distribution function deviates from the true one by $O_{\mathbb{P}}(n^{-1/2})$ uniformly in t, which is the Kolmogorov and Smirnov rate.

Rate of the uniform deviation

Since $\sqrt{n}\Delta_n = \sup_{f \in \mathcal{F}} |\mathbb{G}_n f|$, the maximal inequality with absolute values yields a rate for the uniform deviation of Part III,

$$\mathbb{E}_S[\Delta_n] \leq \frac{C_{\mathrm{MI}}}{\sqrt{n}} \int_0^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{1 + \log N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon)}\, d\epsilon \, .$$

The parametric rate. As soon as the integral is finite, Δ_n is of order $n^{-1/2}$. For instance, a class with $\log N_{[]}(\epsilon) \lesssim p \log(1/\epsilon)$ has an integral of order $\sqrt{p}$, so that Δ_n is of order $\sqrt{p/n}$, and the indicators of the exercise are of this kind.

A question. The integral is finite when the integrand is integrable close to 0, roughly when $\log N_{[]}(\epsilon)$ grows more slowly than ϵ^{-2}. What happens when it is not?

Entropy of Lipschitz functions

To answer the question, we need a class whose entropy grows faster than ϵ^{-2}. The typical one is the nonparametric class $\mathcal{F} = \{f : [0,1]^d \to \mathbb{R}; \|f\|_\infty \le 1, \|f\|_{\mathrm{Lip}} \le 1\}$.

Lemma. There is C_d, depending only on d, such that, for every $\epsilon \in (0,1]$,

$$\log N(\mathcal{F}, \|\cdot\|_\infty, \epsilon) \le C_d \epsilon^{-d} .$$

Proof. We cut $[0,1]^d$ into $K \le (2\sqrt{d}/\epsilon)^d$ cubes of side at most $\epsilon/\sqrt{d}$, with centers $x_1, \dots, x_K$ in lexicographic order, so that each x_j with $j \ge 2$ is adjacent to an earlier center. To $f \in \mathcal{F}$, we associate the function g equal on the j-th cube to the multiple q_j of ϵ closest to $f(x_j)$. Since every point of that cube is within $\epsilon/2$ of x_j, we get $|f - g| \le \epsilon/2 + \epsilon/2 = \epsilon$ there.

It remains to count the functions g. There are at most $4/\epsilon$ choices for q_1. For adjacent cubes, $|q_j - q_{j'}| \le \epsilon/2 + \epsilon + \epsilon/2 = 2\epsilon$, so that each next q_j has at most 5 choices given the earlier ones. Hence, the logarithm of the number of functions g is at most $\log(4/\epsilon) + K \log 5 \le (4 + (2\sqrt{d})^d \log 5)\epsilon^{-d}$, which concludes the proof.

Divergence of the entropy integral

For the Lipschitz class, the lemma bounds the integrand by $\sqrt{1 + C_d \epsilon^{-d}}$, which is integrable at 0 only for $d < 2$. For $d \geq 2$, the maximal inequality thus yields an infinite bound, and we stop the chaining at a scale δ instead.

Proposition (truncated maximal inequality). Let $\mathcal{F}$ have an envelope F with $\mathbb{E}_X[F(X)^2] < \infty$. Then, for every $\delta > 0$ with $4\delta \leq \sqrt{\mathbb{E}_X[F(X)^2]}$,

$$\mathbb{E}^\star[\Delta_n] \leq C \left(\delta + \frac{1}{\sqrt{n}} \int_\delta^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{1 + \log N(\mathcal{F}, \|\cdot\|_\infty, \epsilon)} d\epsilon \right) ,$$

for a universal constant C.

Reading it. The term δ accounts for the scales below δ, where we approximate in supremum norm without chaining, and the integral for the chaining above δ. The cutoff δ is free, and we optimize it.

Polynomial entropy. When $\log N(\mathcal{F}, \|\cdot\|_\infty, \epsilon) \lesssim \epsilon^{-\alpha}$, as for the Lipschitz class with $\alpha = d$, optimizing in δ leads to three regimes, depending on α.

Proof of the truncated maximal inequality

A finite net. Picking one element of $\mathcal{F}$ in each ball of a minimal cover by supremum norm balls of radius 2δ, we get a finite $\mathcal{G} \subset \mathcal{F}$ with $|\mathcal{G}| \leq N(\mathcal{F}, \|\cdot\|_\infty, 2\delta)$ and, for every $f \in \mathcal{F}$, some $g_f \in \mathcal{G}$ such that

$$\|f - g_f\|_\infty \leq 4\delta \,. \quad \text{since } f \text{ and } g_f \text{ lie in one ball of radius } 2\delta$$

Reduction to the net. For every $f \in \mathcal{F}$, by the triangle inequality, we have

$$\underbrace{\left|\frac{1}{n}\sum_{i\leq n} f(X_i) - \mathbb{E}_X[f(X)]\right|}_{\text{supremum over } f \text{ equal to } \Delta_n}$$

$$\leq \left|\frac{1}{n}\sum_{i\leq n} g_f(X_i) - \mathbb{E}_X[g_f(X)]\right| + \underbrace{\left|\frac{1}{n}\sum_{i\leq n}(f-g_f)(X_i)\right|}_{\leq 4\delta} + \underbrace{|\mathbb{E}_X[(f-g_f)(X)]|}_{\leq 4\delta}$$

$$\leq \sup_{g\in\mathcal{G}} \left|\frac{1}{n}\sum_{i\leq n} g(X_i) - \mathbb{E}_X[g(X)]\right| + 8\delta \,. \qquad \text{since } g_f \in \mathcal{G}$$

Proof, the bracketing numbers of the net

Since $\mathcal{G} \subset \mathcal{F}$ has envelope F, the maximal inequality with absolute values applies to $\mathcal{G}$, and the reduction to the net yields

$$\mathbb{E}^\star[\Delta_n] \leq 8\delta + \frac{C_{\mathrm{MI}}}{\sqrt{n}} \underbrace{\int_0^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{1 + \log N_{[]}(\mathcal{G}, L^2(\mathcal{L}), \epsilon)}\, d\epsilon}_{\text{entropy integral of the net}} \ .$$

Balls are brackets. For every function c and every $\epsilon > 0$, we have

$$\{f : \|f - c\|_\infty \leq \epsilon/2\} \subset \underbrace{[c - \epsilon/2, c + \epsilon/2]}_{L^2(\mathcal{L}) \text{ width } \epsilon} \ .$$

Two bounds. Hence, for every $\epsilon > 0$, we have

$$N_{[]}(\mathcal{G}, L^2(\mathcal{L}), \epsilon) \leq N(\mathcal{F}, \|\cdot\|_\infty, 2\delta) \ , \quad \text{with the brackets } [g, g],\ g \in \mathcal{G}, \text{ of width } 0$$
$$N_{[]}(\mathcal{G}, L^2(\mathcal{L}), \epsilon) \leq N(\mathcal{F}, \|\cdot\|_\infty, \epsilon/2) \ . \quad \text{since } \mathcal{G} \subset \mathcal{F} \text{ and balls are brackets}$$

Proof, the entropy integral of the net

We use the first bound for $\epsilon \leq 4\delta$ and the second one above, and we substitute $\epsilon = 2u$ in the last step, which yields

$$
\begin{aligned}
&\int_0^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{1 + \log N_{[]}(\mathcal{G}, L^2(\mathcal{L}), \epsilon)} d\epsilon \\
&\leq \underbrace{4\delta \sqrt{1 + \log N(\mathcal{F}, \|\cdot\|_\infty, 2\delta)}}_{\text{first bound}} + \underbrace{\int_{4\delta}^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{1 + \log N(\mathcal{F}, \|\cdot\|_\infty, \epsilon/2)} d\epsilon}_{\text{second bound}} \\
&\leq \underbrace{2 \int_{2\delta}^{4\delta} \sqrt{1 + \log N(\mathcal{F}, \|\cdot\|_\infty, \epsilon/2)} d\epsilon}_{N(\mathcal{F}, \|\cdot\|_\infty, \epsilon/2) \geq N(\mathcal{F}, \|\cdot\|_\infty, 2\delta) \text{ on } [2\delta, 4\delta]} + \int_{4\delta}^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{1 + \log N(\mathcal{F}, \|\cdot\|_\infty, \epsilon/2)} d\epsilon \\
&\leq 4 \int_{\delta}^{\sqrt{\mathbb{E}_X[F(X)^2]}} \sqrt{1 + \log N(\mathcal{F}, \|\cdot\|_\infty, u)} du \ .
\end{aligned}
$$

Substituting into the maximal inequality on the net yields the result with $C = \max(8, 4C_{\text{MI}})$, which concludes the proof.

Nonparametric rates, the three regimes

Let $\mathcal{F}$ have envelope 1 and satisfy $\log N(\mathcal{F}, \|\cdot\|_\infty, \epsilon) \lesssim \epsilon^{-\alpha}$ for some $\alpha > 0$. The truncated maximal inequality yields $\mathbb{E}^\star[\Delta_n] \lesssim \delta + n^{-1/2} \int_\delta^1 \epsilon^{-\alpha/2} d\epsilon$, and optimizing in δ leads to

entropy exponent	$\int_\delta^1 \epsilon^{-\alpha/2} d\epsilon$	best δ	rate of $\mathbb{E}^\star[\Delta_n]$
$\alpha < 2$	converges at 0	0	$n^{-1/2}$
$\alpha = 2$	$\log(1/\delta)$	$n^{-1/2}$	$n^{-1/2} \log n$
$\alpha > 2$	$\delta^{1-\alpha/2}$	$n^{-1/\alpha}$	$n^{-1/\alpha}$

Reading the table. Below $\alpha = 2$, we recover the parametric rate. Above, the rate $n^{-1/\alpha}$ is slower, and the larger the class, the slower the rate.

A rate for the value of the objective

The maximal inequality already yields a rate for the value of the objective, i.e. for $\sup_{\theta\in\Theta} M_n(\theta)$ as an estimator of $\sup_{\theta\in\Theta} M(\theta)$. Indeed, since $|M_n - M| \le \Delta_n$ on Θ, the two suprema differ by at most Δ_n, and the value inherits the rate of the uniform deviation,

$$\mathbb{E}_S\left[\left|\sup_{\theta\in\Theta} M_n(\theta) - \sup_{\theta\in\Theta} M(\theta)\right|\right] \le \mathbb{E}_S[\Delta_n]\ .$$

No assumption on Θ. Neither a unique maximizer, nor identifiability, nor smoothness is needed, and the statement says nothing about $\hat\theta_n$.

Every rate transfers. The value converges at rate $n^{-1/2}$ for a parametric class or for the indicators, and at rate $n^{-1/d}$ for the Lipschitz class in dimension $d \ge 3$.

From a rate to a limit law. A rate gives the order of the error, not its law. A gaussian limit at the $\sqrt{n}$ scale requires more, as the next slide explains.

The empirical process at the estimator

The two limit laws to come, for the value of the objective and for the parameter, both need the limit law of the empirical process at the estimator. With $m_\theta = m(\cdot, \theta)$, $M_n(\theta) = \frac{1}{n}\sum_{i \le n} m(X_i, \theta)$ and $M(\theta) = \mathbb{E}_X[m(X, \theta)]$, it is

$$\mathbb{G}_n m_{\hat{\theta}_n} = \sqrt{n}\left(M_n(\hat{\theta}_n) - M(\hat{\theta}_n)\right) \ ,$$

and $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta^\star$, in one of two roles:

- **for the value**, m is the criterion and $\theta^\star$ the maximizer of M,
- **for the parameter**, $m(x, \theta) = z(x, \theta)_j$ is the j-th coordinate of the estimating function, for each $j \le p$, and $\theta^\star = \theta_0$.

At a fixed point. For a fixed θ, $\mathbb{G}_n m_\theta$ is a normalized sum of i.i.d. centered terms, and the central limit theorem yields $\mathbb{G}_n m_\theta \xrightarrow{d} \mathcal{N}(0, \mathrm{Var}_X(m(X, \theta)))$.

At the estimator. Since $\hat{\theta}_n$ depends on the sample, $\mathbb{G}_n m_{\hat{\theta}_n}$ is not such a sum, and the central limit theorem does not apply. This is the step hidden in the $\approx$ of the heuristic.

Localisation of the empirical process

Since $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta^\star$, we split

$$\mathbb{G}_n m_{\hat{\theta}_n} = \underbrace{\mathbb{G}_n m_{\theta^\star}}_{\text{central limit theorem}} + \underbrace{\mathbb{G}_n(m_{\hat{\theta}_n} - m_{\theta^\star})}_{\text{remainder}},$$

and we bound the remainder by its supremum over a small ball around $\theta^\star$, which the maximal inequality controls.

The localisation lemma

Lemma. Let $\theta^\star \in \Theta$, let A be a neighbourhood of $\theta^\star$, let $m_\theta = m(\cdot, \theta)$ and assume

1. $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta^\star$,
2. $\mathcal{F} = \{m_\theta ; \theta \in A\}$ satisfies $\int_0^\delta \sqrt{\log N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon)} d\epsilon < \infty$ for every δ,
3. $\mathbb{E}_X[F_\delta(X)^2] \to 0$ as $\delta \to 0$, where $F_\delta(x) = \sup_{\theta \in A, \|\theta - \theta^\star\| \leq \delta} |m(x, \theta) - m(x, \theta^\star)|$.

Then $\mathbb{G}_n \left(m_{\hat{\theta}_n} - m_{\theta^\star} \right) = o_{\mathbb{P}}(1)$.

Reading the lemma. The remainder of the localisation vanishes, so that $\mathbb{G}_n m_{\hat{\theta}_n} = \mathbb{G}_n m_{\theta^\star} + o_{\mathbb{P}}(1)$, and the central limit theorem applies to the fixed point $\theta^\star$.

Role of each hypothesis. Hypothesis 1 puts $\hat{\theta}_n$ in a small ball, hypothesis 3 makes the increments $m_\theta - m_{\theta^\star}$ small in L^2 on that ball, and hypothesis 2 allows the maximal inequality to turn this into a small supremum.

Proof of the localisation lemma

For $\delta > 0$ with $B(\theta^\star, \delta) \subset A$, let $\mathcal{F}_\delta = \{m_\theta - m_{\theta^\star} ; \|\theta - \theta^\star\| \leq \delta\}$, which has envelope F_δ. Since subtracting the fixed function $m_{\theta^\star}$ from both ends of a bracket leaves its width unchanged, we have

$$N_{[]}(\mathcal{F}_\delta, L^2(\mathcal{L}), \epsilon) \leq N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon) \ .$$

On the event $\{\|\hat{\theta}_n - \theta^\star\| \leq \delta\}$, whose probability tends to 1 by hypothesis 1, the random argument belongs to $\mathcal{F}_\delta$. Hence, by Markov's inequality and the maximal inequality, for every ϵ and every such δ,

$$\begin{aligned}\limsup_n \mathbb{P}\left(\left|\mathbb{G}_n(m_{\hat{\theta}_n} - m_{\theta^\star})\right| \geq \epsilon\right) &\leq \limsup_n \mathbb{P}\left(\sup_{f \in \mathcal{F}_\delta} |\mathbb{G}_n f| \geq \epsilon\right) \\ &\leq \frac{C_{\text{MI}}}{\epsilon} \int_0^{\sqrt{\mathbb{E}_X[F_\delta(X)^2]}} \sqrt{1 + \log N_{[]}(\mathcal{F}, L^2(\mathcal{L}), u)}\, du \ .\end{aligned}$$

The left side does not depend on δ. Hypothesis 3 sends the upper limit of the integral to 0, hypothesis 2 makes the integrand integrable there, and the 1 contributes no more than that same upper limit, so the right side is arbitrarily small for δ small enough, which proves that the left side is 0.

A limit law for the value

Theorem. Let $\theta^\star$ be a well separated maximizer of M, let $\hat{\theta}_n$ maximize M_n up to $o_{\mathbb{P}}(n^{-1/2})$, and assume the hypotheses of the localisation lemma at $\theta^\star$. Then

$$\sqrt{n}\left(\sup_{\theta\in\Theta} M_n(\theta) - \sup_{\theta\in\Theta} M(\theta)\right) \xrightarrow{d} \mathcal{N}(0, \mathrm{Var}_X(m(X,\theta^\star))) \,.$$

Proof, the lower bound. Since $\sup_\Theta M = M(\theta^\star)$, taking $\theta = \theta^\star$ in the supremum on the left yields

$$\begin{aligned}\sqrt{n}\left(\sup_{\Theta} M_n - \sup_{\Theta} M\right) &\geq \sqrt{n}\,(M_n(\theta^\star) - M(\theta^\star)) \\ &= \mathbb{G}_n m_{\theta^\star} \,.\end{aligned}$$

Proof, the upper bound

Since $\sup_\Theta M_n = M_n(\hat\theta_n) + o_{\mathbb{P}}(n^{-1/2})$ and $M(\hat\theta_n) \leq M(\theta^\star)$, we have

$$\begin{aligned}\sqrt{n}\left(\sup_\Theta M_n - \sup_\Theta M\right) &= \mathbb{G}_n m_{\hat\theta_n} + \sqrt{n}\left(M(\hat\theta_n) - M(\theta^\star)\right) + o_{\mathbb{P}}(1) \\ &\leq \mathbb{G}_n m_{\theta^\star} + \mathbb{G}_n\left(m_{\hat\theta_n} - m_{\theta^\star}\right) + o_{\mathbb{P}}(1)\ .\end{aligned}$$

The last term is $o_{\mathbb{P}}(1)$ by the localisation lemma, so that the upper and the lower bounds both equal $\mathbb{G}_n m_{\theta^\star} + o_{\mathbb{P}}(1)$, and the central limit theorem concludes the proof.

Tools of the proof. We used no weak convergence in a space of functions, no argmax set and no differentiability of a supremum. The only tool beyond the central limit theorem is the lemma, and the only hypothesis beyond consistency is that the maximizer is unique and well separated.

Example, empirical risk minimization

If the risk minimizer $f^\star$ is unique in the class, the theorem applied to $M_n = -\widehat{R}_S$ yields

$$\sqrt{n}\left(\widehat{R}_S(\hat{f}_S) - \inf_{\theta\in\Theta} R(f_\theta)\right) \xrightarrow{d} \mathcal{N}\left(0, \mathrm{Var}_{(x,y)}(\ell(y, f^\star(x)))\right) .$$

A limit law for the parameter, hypotheses

We now turn to the parameter of a Z-estimator, and we make the heuristic rigorous.

Theorem. Let $X_1, X_2, \dots$ be i.i.d. of law $\mathcal{L}$ on $\mathbb{R}^d$ and $z : \mathbb{R}^d \times \Theta \to \mathbb{R}^p$ with $\mathbb{E}_X[\|z(X,\theta)\|^2] < \infty$ for every θ. Assume

1. $\frac{1}{\sqrt{n}} \sum_{i \leq n} z(X_i, \hat{\theta}_n) = o_{\mathbb{P}}(1)$, and there is θ_0 in the interior of Θ with $\mathbb{E}_X[z(X,\theta_0)] = 0$ and $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$,
2. on a neighbourhood A of θ_0, $\theta \mapsto \mathbb{E}_X[z(X,\theta)]$ is continuously differentiable, with Jacobian V at θ_0 invertible,
3. for $j \leq p$, the class $\mathcal{F}_j = \{x \mapsto z(x,\theta)_j ; \theta \in A\}$ satisfies $\int_0^\delta \sqrt{\log N_{[]}(\mathcal{F}_j, L^2(\mathcal{L}), \epsilon)} d\epsilon < \infty$ for every δ,
4. $\mathbb{E}_X \left[\sup_{\theta \in A, \|\theta - \theta_0\| \leq \delta} \|z(X,\theta) - z(X,\theta_0)\|^2 \right] \xrightarrow[\delta \to 0]{} 0.$

A limit law for the parameter, conclusion

Theorem, continued. Under those four hypotheses, the estimator admits a linear expansion,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V^{-1} \frac{1}{\sqrt{n}} \sum_{i \leq n} z(X_i, \theta_0) + o_{\mathbb{P}}(1) \ ,$$

and thus

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, V^{-1} \mathrm{Cov}_X(z(X, \theta_0)) V^{-\top}) \ .$$

The expansion is the stronger statement. It writes the error as an average of i.i.d. terms plus a negligible remainder, so that the limit law follows from the central limit theorem.

Role of each hypothesis

Hypothesis 1 is the input. It is the conclusion of Parts II and III, together with the estimating equation itself. Nothing in Part IV proves consistency, which is assumed and used to localise.

Hypothesis 2 is the only smoothness, and it is on the expectation. It allows for a discontinuous z, and the matrix V of the limit comes from it. Note that it is deterministic.

Hypothesis 3 is the size of the class, which the maximal inequality requires. It is probabilistic in use and combinatorial in nature.

Hypothesis 4 is an L^2 continuity of z at θ_0, which makes the envelope of the localised class shrink as the neighbourhood shrinks. Without it, the maximal inequality yields a bound that does not tend to 0.

Proof, expanding the expectation

We denote $G(\theta) = \mathbb{E}_X[z(X, \theta)]$ and $V = JG(\theta_0)$. Hypothesis 2 and differentiability at θ_0 yield, as $\theta \to \theta_0$,

$$\begin{aligned} G(\theta) &= G(\theta_0) + V(\theta - \theta_0) + o(\|\theta - \theta_0\|) \\ &= V(\theta - \theta_0) + o(\|\theta - \theta_0\|)\ . \qquad\qquad \text{since } G(\theta_0) = 0 \end{aligned}$$

Since $\hat{\theta}_n - \theta_0 = o_{\mathbb{P}}(1)$ by hypothesis 1, the transfer lemma of lecture 2 allows us to evaluate it at the random point:

$$\begin{aligned} G(\hat{\theta}_n) &= V(\hat{\theta}_n - \theta_0) + o_{\mathbb{P}}(\|\hat{\theta}_n - \theta_0\|) \\ &= (V + Q_n)(\hat{\theta}_n - \theta_0)\ , \end{aligned}$$

where $\|Q_n\| = o_{\mathbb{P}}(1)$.

The remainder as a matrix. Let ρ_n be the remainder and $h_n = \hat{\theta}_n - \theta_0$. We set $Q_n = \rho_n h_n^\top / \|h_n\|^2$ when $h_n \neq 0$ and $Q_n = 0$ otherwise, so that $Q_n h_n = \rho_n$ and $\|Q_n\| = \|\rho_n\| / \|h_n\| = o_{\mathbb{P}}(1)$.

Proof, multiplying by the rate

Multiplying by $\sqrt{n}$ and using hypothesis 1 in the form $\frac{1}{\sqrt{n}} \sum_{i \leq n} z(X_i, \hat{\theta}_n) = o_{\mathbb{P}}(1)$, we obtain

$$\begin{aligned}(V + Q_n)\sqrt{n}(\hat{\theta}_n - \theta_0) &= \sqrt{n} G(\hat{\theta}_n) \\ &= \frac{1}{\sqrt{n}} \sum_{i \leq n} \Big(G(\hat{\theta}_n) - z(X_i, \hat{\theta}_n) \Big) + o_{\mathbb{P}}(1) \ .\end{aligned}$$

Adding and subtracting $z(X_i, \theta_0)$ and $G(\theta_0) = 0$ in the right side, we obtain

$$(V + Q_n)\sqrt{n}(\hat{\theta}_n - \theta_0) = o_{\mathbb{P}}(1) - \frac{1}{\sqrt{n}} \sum_{i \leq n} z(X_i, \theta_0) + r_n \ ,$$

with

$$r_n = \frac{1}{\sqrt{n}} \sum_{i \leq n} \Big\{ \Big(z(X_i, \theta_0) - z(X_i, \hat{\theta}_n) \Big) - \Big(G(\theta_0) - G(\hat{\theta}_n) \Big) \Big\} \ .$$

By the central limit theorem, the middle term converges in law to $\mathcal{N}(0, \mathrm{Cov}_X(z(X, \theta_0)))$, and it remains to show that $r_n = o_{\mathbb{P}}(1)$.

Proof, conclusion

The remainder falls under the localisation lemma, applied to each coordinate. Denoting $z_\theta = z(\cdot, \theta)$, we have

$$r_n = -\mathbb{G}_n \left(z_{\hat{\theta}_n} - z_{\theta_0} \right) \ ,$$

and hypotheses 1, 3 and 4 of the theorem are hypotheses 1, 2 and 3 of the lemma. Hence, $r_n = o_{\mathbb{P}}(1)$.

It remains to invert. We have obtained

$$(V + Q_n)\sqrt{n}(\hat{\theta}_n - \theta_0) = -\frac{1}{\sqrt{n}} \sum_{i \leq n} z(X_i, \theta_0) + o_{\mathbb{P}}(1) \ ,$$

with $\|Q_n\| = o_{\mathbb{P}}(1)$ and V invertible, so that $V + Q_n$ is invertible with probability going to 1 and $(V + Q_n)^{-1} \xrightarrow{\mathbb{P}} V^{-1}$. Slutsky's lemma yields both conclusions of the theorem.

The empirical median, hypotheses 1 and 2

Let $X_1, X_2, \ldots$ be i.i.d. with distribution function F_X and density f, strictly positive on $\mathbb{R}$, and let $\hat{\theta}_n$ be the empirical median, so that $z(x, \theta) = \operatorname{sign}(\theta - x)$ and $\frac{1}{n}\sum_{i \le n} z(X_i, \hat{\theta}_n) = 0$.

Hypothesis 1. $\mathbb{E}_X[z(X, \theta)] = 2F_X(\theta) - 1$ vanishes at the population median θ_0, and $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$ by the monotone theorem of Part II.

Hypothesis 2. If f is continuous near θ_0, then $\theta \mapsto 2F_X(\theta) - 1$ is continuously differentiable there, with derivative $V = 2f(\theta_0) > 0$, which is invertible as a 1×1 matrix.

It remains to check hypotheses 3 and 4, and hypothesis 3 requires some work, since it involves the bracketing numbers of the class of all sign functions.

Bracketing the sign functions

Lemma. Let $\mathcal{L}$ be an atomless law on $\mathbb{R}$ and $\mathcal{F} = \{x \mapsto \operatorname{sign}(\theta - x); \theta \in \mathbb{R}\}$. Then for every $\epsilon > 0$,

$$N_{[]}\left(\mathcal{F}, L^2(\mathcal{L}), \epsilon\right) \leq \frac{4}{\epsilon^2} + 1 \, .$$

Proof, the grid. Let N be the smallest integer with $N + 1 \geq \epsilon^{-2}$, and let us choose $t_0 = -\infty < t_1 < \cdots < t_N < t_{N+1} = +\infty$ with $\mathcal{L}((t_j, t_{j+1})) \leq \epsilon^2$ for every j, which is possible by taking quantiles since $\mathcal{L}$ has no atoms.

Proof, two half classes. For $j \leq N$, we set $l_{+,j}(x) = \mathbf{1}\{x \leq t_j\}, u_{+,j}(x) = \mathbf{1}\{x < t_{j+1}\}, l_{-,j}(x) = \mathbf{1}\{t_{j+1} < x\}$ and $u_{-,j}(x) = \mathbf{1}\{t_j < x\}$. For $t_j < \theta \leq t_{j+1}$, we have

$$\mathbf{1}\{\cdot < \theta\} \in [l_{+,j}, u_{+,j}] \, ,$$
$$\mathbf{1}\{\theta < \cdot\} \in [l_{-,j}, u_{-,j}] \, ,$$

and both differences equal $\mathbf{1}\{t_j < x < t_{j+1}\}$ up to an $\mathcal{L}$-null set, of $L^2(\mathcal{L})$ norm $\sqrt{\mathcal{L}((t_j, t_{j+1}))} \leq \epsilon$.

Bracketing the sign functions, assembly

Since $\operatorname{sign}(\theta - x) = \mathbf{1}\{x < \theta\} - \mathbf{1}\{\theta < x\}$, the two displays combine, for $t_j < \theta \le t_{j+1}$, into

$$\operatorname{sign}(\theta - \cdot) \in [l_{+,j} - u_{-,j},\, u_{+,j} - l_{-,j}] \,.$$

By the triangle inequality, the width of that bracket satisfies

$$\begin{aligned} \|(u_{+,j} - l_{-,j}) - (l_{+,j} - u_{-,j})\|_{L^2(\mathcal{L})} &= \|(u_{+,j} - l_{+,j}) + (u_{-,j} - l_{-,j})\|_{L^2(\mathcal{L})} \\ &\le 2\epsilon \,. \end{aligned}$$

These $N + 1$ brackets cover $\mathcal{F}$, so that $N_{[]}(\mathcal{F}, L^2(\mathcal{L}), 2\epsilon) \le N + 1 \le \epsilon^{-2} + 1$, and replacing ϵ by $\epsilon/2$ yields the statement. In particular, $\log N_{[]}(\epsilon) \lesssim \log(1/\epsilon)$, and the entropy integral of hypothesis 3 converges.

The empirical median, the limit law

Hypothesis 4. For $|\theta - \theta_0| \leq \delta$, $\operatorname{sign}(\theta - X)$ and $\operatorname{sign}(\theta_0 - X)$ differ only when $X \in [\theta_0 - \delta, \theta_0 + \delta]$, and then by ± 2. Hence,

$$\mathbb{E}_X \left[\sup_{|\theta - \theta_0| \leq \delta} \left(\operatorname{sign}(\theta - X) - \operatorname{sign}(\theta_0 - X) \right)^2 \right] = 4\mathbb{P}\left(X \in [\theta_0 - \delta, \theta_0 + \delta] \right) \xrightarrow[\delta \to 0]{} 0 \, .$$

The limit. Since $\operatorname{Cov}_X(z(X, \theta_0)) = \mathbb{E}_X[\operatorname{sign}(\theta_0 - X)^2] = 1$ and $V = 2f(\theta_0)$, the sandwich yields

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{4f(\theta_0)^2} \right) \, .$$

Part V. Analysis of maximum likelihood

Maximum likelihood, the results so far

Definition (recalled). With $X_1, \dots, X_n$ i.i.d. of law $\mathcal{L}_{\theta_0}$ and f_θ the density of $\mathcal{L}_\theta$, the maximum likelihood estimator $\hat{\theta}_n$ is the M-estimator associated with $M_n(\theta) = \frac{1}{n}\sum_{i\leq n} \log f_\theta(X_i)$.

proved in	statement for maximum likelihood	under
Part III	$\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$	items 1 to 5, including identifiability
Part III	$\mathrm{KL}(\mathcal{L}_{\theta_0} \| \mathcal{L}_{\hat{\theta}_n}) \xrightarrow{\mathbb{P}} 0$	$\{\log f_\theta ; \theta \in \Theta\}$ Glivenko-Cantelli

The tool of Part IV. Any Z-estimator whose estimating function z satisfies hypotheses 1 to 4 of the limit law for the parameter is asymptotically normal, with covariance $V^{-1}\mathrm{Cov}_X(z(X,\theta_0))V^{-\top}$.

The remaining step. The two results above locate the parameter and the law, but not the size of the error $\hat{\theta}_n - \theta_0$. It remains to write maximum likelihood as a Z-estimator with the score $\nabla_\theta \log f_\theta$, to check hypotheses 1 to 4 for it, and to compute V, in order to prove that, with $I(\theta_0)$ the Fisher information matrix,

Bracketing of Lipschitz parametrized classes

To check hypothesis 3 for maximum likelihood, we need the bracketing numbers of a class indexed by a parameter of finite dimension, which the following lemma provides.

Lemma. Let $\mathcal{L}$ be a law on $\mathbb{R}^d$, let $\Theta \subset \mathbb{R}^p$ be bounded, and let $\mathcal{F} = \{g_\theta ; \theta \in \Theta\} \subset L^2(\mathcal{L})$. Assume that there is a nonnegative $h \in L^2(\mathcal{L})$ such that, for all $\theta_1, \theta_2 \in \Theta$ and $x \in \mathbb{R}^d$,

$$|g_{\theta_1}(x) - g_{\theta_2}(x)| \leq \|\theta_1 - \theta_2\| h(x) \ .$$

Then there is C_p, depending only on p, such that, for every $\epsilon > 0$,

$$N_{[]}(\mathcal{F}, L^2(\mathcal{L}), \epsilon) \leq \max\left(1, C_p \left(\frac{\operatorname{diam}(\Theta) \|h\|_{L^2(\mathcal{L})}}{\epsilon}\right)^p\right) \ .$$

Consequence. The logarithm of the bracketing numbers grows like $p \log(1/\epsilon)$ as $\epsilon \to 0$, so that the entropy integral is finite, as for the indicators of the exercise.

Proof of the bracketing lemma

A net of Θ. Let $\delta \in (0, \operatorname{diam}(\Theta)]$. We cut a cube of side $2\operatorname{diam}(\Theta)$ containing Θ into cubes of side $\delta/\sqrt{p}$, hence of diameter δ, and we pick one point of Θ in each small cube meeting Θ. Every $\theta \in \Theta$ is then within δ of one of these points $\theta_1, \dots, \theta_N$, and

$$\begin{aligned} N &\leq \left\lceil \frac{2\sqrt{p}\operatorname{diam}(\Theta)}{\delta} \right\rceil^p \\ &\leq \left(\frac{3\sqrt{p}\operatorname{diam}(\Theta)}{\delta} \right)^p . \quad \text{since } \lceil u \rceil \leq 3u/2 \text{ for } u \geq 2 \end{aligned}$$

Brackets. If $\|\theta - \theta_j\| \leq \delta$, the Lipschitz hypothesis yields $|g_\theta - g_{\theta_j}| \leq \delta h$, so that g_θ lies in

$$\underbrace{[g_{\theta_j} - \delta h, g_{\theta_j} + \delta h]}_{L^2(\mathcal{L}) \text{ width } 2\delta \|h\|_{L^2(\mathcal{L})}} .$$

Conclusion. These N brackets cover $\mathcal{F}$. For $\epsilon \leq 2\operatorname{diam}(\Theta)\|h\|_{L^2(\mathcal{L})}$, the choice $\delta = \epsilon/(2\|h\|_{L^2(\mathcal{L})})$ yields the bound with $C_p = (6\sqrt{p})^p$. For larger ϵ, every θ is within $\operatorname{diam}(\Theta)$ of θ_1, so that the single bracket $[g_{\theta_1} - \operatorname{diam}(\Theta)h, g_{\theta_1} + \operatorname{diam}(\Theta)h]$ suffices, which concludes the proof.

Asymptotic normality of maximum likelihood

Theorem. Let $X_1, X_2, \dots$ be i.i.d. with density f_{θ_0} with respect to Lebesgue measure, with θ_0 in the interior of Θ, and let $A \subset \Theta$ be a closed ball centered at θ_0. We denote by $z(x, \theta) = \nabla_\theta \log f_\theta(x)$ the **score**, by ∂_a the derivative in θ_a, and we assume, for $a, b \leq p$,

- **regularity**: for every x, $\theta \mapsto f_\theta(x)$ is positive and C^2, and $\mathbb{E}_X[\|z(X, \theta)\|^2] < \infty$ for every θ,
- **consistency**: $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$, for instance under the hypotheses of Part III,
- **domination of the density**: $\sup_{\theta \in A} \|\nabla_\theta f_\theta\|$ and $\sup_{\theta \in A} |\partial_a \partial_b f_\theta|$ are Lebesgue integrable,
- **domination of the log-density**: $\mathbb{E}_X[\sup_{\theta \in A} (\partial_a \partial_b \log f_\theta(X))^2] < \infty$,
- **invertibility**: the **Fisher information matrix** $I(\theta_0) = \mathrm{Cov}_X(z(X, \theta_0))$ is invertible.

Then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}) \, .$$

Proof, hypothesis 1

We apply the limit law for Z-estimators to the score, and we check its four hypotheses in turn.

The estimating equation. Since $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$ with θ_0 interior, $\hat{\theta}_n$ lies in the interior of A with probability tending to 1. On that event, it maximizes the differentiable function M_n at an interior point, so that $\nabla M_n(\hat{\theta}_n) = \frac{1}{n}\sum_{i\leq n} z(X_i, \hat{\theta}_n) = 0$. Hence, $\frac{1}{\sqrt{n}}\sum_{i\leq n} z(X_i, \hat{\theta}_n) = o_{\mathbb{P}}(1)$.

The centered score. We have

$$\begin{aligned}
\mathbb{E}_X[z(X, \theta_0)] &= \int \frac{\nabla f_{\theta_0}(x)}{f_{\theta_0}(x)} f_{\theta_0}(x) dx \\
&= \nabla \int f_{\theta_0}(x) dx && \text{by domination of the density} \\
&= 0\,, && \text{since } \textstyle\int f_\theta = 1 \text{ for every } \theta
\end{aligned}$$

which, with consistency, concludes the proof of hypothesis 1.

Proof, hypothesis 2

Smoothness. The domination of the log-density bounds $\mathbb{E}_X[\sup_{\theta\in A}|\partial_a\partial_b \log f_\theta(X)|]$. Hence, we may differentiate $\theta \mapsto \mathbb{E}_X[z(X,\theta)]$ under the integral sign, and by dominated convergence, it is C^1 on A, with Jacobian $\mathbb{E}_X[\nabla^2_\theta \log f_\theta(X)]$.

The Jacobian at θ_0. Using $\partial_a\partial_b \log f = \frac{\partial_a\partial_b f}{f} - \frac{\partial_a f \partial_b f}{f^2}$, we obtain

$$\begin{aligned} V_{a,b} &= \mathbb{E}_X[\partial_a\partial_b \log f_{\theta_0}(X)] \\ &= \int \partial_a\partial_b f_{\theta_0}(x)dx - I(\theta_0)_{a,b} \quad \text{since } I(\theta_0) = \mathbb{E}_X[z(X,\theta_0)z(X,\theta_0)^\top] \\ &= -I(\theta_0)_{a,b}\,. \qquad\qquad\qquad\qquad\qquad\qquad \text{since } \textstyle\int \partial_a\partial_b f_{\theta_0} = \partial_a\partial_b \int f_{\theta_0} = 0 \end{aligned}$$

Note that the second line uses that the score is centered, and the third the domination of the density. Hence, $V = -I(\theta_0)$ is invertible, which concludes the proof of hypothesis 2.

Proof, hypothesis 3

A Lipschitz score. Since A is convex, the mean value inequality yields, for $\theta_1, \theta_2 \in A$ and $j \le p$,

$$\begin{aligned} |z(x,\theta_1)_j - z(x,\theta_2)_j| &\le \|\theta_1 - \theta_2\| \sup_{\theta \in A} \|\nabla_\theta \partial_j \log f_\theta(x)\| \\ &\le \|\theta_1 - \theta_2\| \underbrace{\sqrt{p} \max_{a,b \le p} \sup_{\theta \in A} |\partial_a \partial_b \log f_\theta(x)|}_{=: h(x)} , \end{aligned}$$

where $h \in L^2(\mathcal{L}_{\theta_0})$ by domination of the log-density.

Hypothesis 3. The bracketing lemma, applied to $\mathcal{F}_j = \{z(\cdot,\theta)_j ; \theta \in A\}$ with this h, yields $\log N_{[]}(\mathcal{F}_j, L^2(\mathcal{L}_{\theta_0}), \epsilon) \le \log C_p + p \log(\operatorname{diam}(A) \|h\|_{L^2(\mathcal{L}_{\theta_0})} / \epsilon)$ for small ϵ, whose square root is integrable at 0.

Proof, hypothesis 4

For δ small enough, the ball of radius δ around θ_0 lies in A, and the mean value inequality on its segments yields

$$
\begin{aligned}
&\mathbb{E}_X\left[\sup_{\|\theta-\theta_0\|\le\delta}\|z(X,\theta)-z(X,\theta_0)\|^2\right] \\
&\le \delta^2\mathbb{E}_X\left[\sup_{\theta\in A}\|\nabla_\theta^2\log f_\theta(X)\|^2\right] \\
&\le \delta^2\sum_{a,b\le p}\mathbb{E}_X\left[\sup_{\theta\in A}(\partial_a\partial_b\log f_\theta(X))^2\right] && \text{since the operator norm is at most the Frobenius norm} \\
&\xrightarrow[\delta\to 0]{} 0\ , && \text{by domination of the log-density}
\end{aligned}
$$

which is hypothesis 4.

Proof, collapse of the sandwich

The four hypotheses hold, so that the limit law for Z-estimators applies. Since $V = -I(\theta_0)$ is symmetric, its covariance collapses,

$$\begin{aligned} V^{-1}\mathrm{Cov}_X(z(X,\theta_0))V^{-\top} &= I(\theta_0)^{-1}I(\theta_0)I(\theta_0)^{-1} \\ &= I(\theta_0)^{-1} \ , \end{aligned}$$

which concludes the proof,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}) \ .$$